Chapter 9 One-Pager — Multi-Agent Reinforcement Learning

Research on the possibilities for applying Artificial Intelligence in computer games

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July 2026

Problem. Chapters 2–8 lived inside two-player zero-sum games, where a Nash strategy secures a value `v*` against anyone and CFR provably converges to it. Chapter 9 is the pivot into the multi-agent world, where the defining difficulty is **non-stationarity**: every agent is learning at once, so from any one agent's seat the environment (which contains the others) never holds still. This is the launch point for the thesis's multi-agent contributions — dynamic opponent modeling (#1), safe exploitation without a minimax anchor (#2), and population-level evaluation (#3).

Approach. Build and compare the field's four structural answers to non-stationarity on small, *exactly-solvable* testbeds, so every learner is graded against ground truth. **Independent learning** (the control that fails) on four canonical matrix games; **CTDE** — centralize the critic at training, decentralize the actor at execution — via MADDPG and MAPPO on cooperative tasks; **PSRO** — a game played over a *population* of policies (meta-Nash + best-response oracle), reusing Chapter 07's exact best response as both oracle and exploitability metric — on Kuhn, Leduc, a matrix game, and a native Goofspiel; **learned communication** (CommNet); and **LOLA** (differentiate through the opponent's learning step) on the Iterated Prisoner's Dilemma. **All numbers below are measured.**

Key results (measured).

- *Independent learning fails exactly where theory says.* It converges to Nash on Prisoner's Dilemma (defection), Stag Hunt (the risk-dominant corner), and Battle of the Sexes, but **Matching Pennies never converges** — its distance-to-Nash actually *grows* (0.30 → 0.48), a sharper-than-predicted non-convergence (a contradicted prediction, §9 of the report).
- *PSRO is the game-theory↔MARL bridge, and it works on the small games.* Exploitability → machine zero on **Kuhn** (0.917 → 2×10^{-16}, 6 rounds), → 0 on the matrix game and Rock–Paper–Scissors. Self-play confirms the mechanism: on Kuhn the **average**-iterate NashConv falls 0.24 → 0.031 while the **last** iterate keeps oscillating — why a population/averaging is needed.
- *A centralized critic is a near-zero-variance teacher.* On CoopSignal the centralized critic's residual is `3.2e-11` vs the independent critic's `0.077` — the CTDE variance-reduction claim, confirmed.
- *Communication clears the guessing ceiling.* CommNet with the channel ON scores `0.795` vs `0.204` OFF (ceiling 1/K = 0.2) — a learned, not designed, protocol.
- *LOLA turns defectors into cooperators.* IPD per-step return rises from `1.04` (naive defection) to `2.82` (LOLA cooperation); zeroing the look-ahead recovers the naive gradient exactly (the mechanism check).
- *Honest negatives (kept predictions, §9).* PSRO on **Leduc** declines but hits a scaling wall (`4.75→2.16` after 20 rounds, not `<0.5`); **Goofspiel K=4** oscillates rather than converging (a flagged code anomaly, documented not fixed); and on the **climbing game** no method reaches the optimum 11 (IL/MAPPO 7, MADDPG 5) — a centralized critic lowers variance but does not by itself solve hard-exploration coordination.

Thesis connection. PSRO is Chapter 2's iterated best response lifted to a population and the empirical backbone here; LOLA is *dynamic* opponent modeling, the moving-target complement to Chapter 7's static read (Contribution #1); PSRO's meta-game is a population-level evaluation methodology (Contribution #3). The Leduc scaling wall echoes Chapter 8's global-vs-local finding, and the place where every two-player guarantee stops — the **missing `N>2` minimax anchor** — is exactly Contribution #2's problem statement.

Open questions. Can an approximate (RL) oracle push Leduc PSRO close to Nash, and at what cost to the guarantee? What breaks the Goofspiel-K=4 oscillation (de-duplication? a mixed population? a different meta-solver?), and why does a centralized critic *hurt* on the climbing game? And underneath all of it: with no minimax value for `N>2`, what does "safe" mean for a population, and can PSRO's meta-game supply the substitute (Contribution #2)?